Matrix Representation of Linear Transformation :-

$T : U \to V$

Let S and S' are the basis of U and V respectively.

$S = \{u_1, \dots, u_m\}$

$S' = \{v_1, v_2, \dots, v_n\}$

$[T(u_1)]_{S'}, [T(u_2)]_{S'}, \dots, [T(u_m)]_{S'}$

$[T(u_1)]_{S'} = [a_{11}, a_{21}, \dots, a_{n1}]$

$[T(u_2)]_{S'} = [a_{12}, a_{22}, \dots, a_{n2}]$

$\vdots$

$[T(u_n)]_{S'} = [a_{1m}, a_{2m}, \dots, a_{nm}]$

$$(T)_{SS'} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}_{n \times m}$$

Ex:- $T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(x, y, z) = (x + y, y - z)$

$S = \{(1,0,0), (0,1,0), (0,0,1)\}$

$S' = \{(1,0), (0,1)\}$

$T(1,0,0) = (1,0) = 1(1,0) + 0(0,1)$ $\quad [(1,0)]_{S'} = (1,0)^T$

$T(0,1,0) = (1,1) = 1(1,0) + 1(0,1)$ $\quad [(1,1)]_{S'} = (1,1)^T$

$T(0,0,1) = (0,-1) = 0(1,0) - 1(0,1)$ $\quad [(0,-1)]_{S'} = (0,-1)^T$

$$(T)_{SS'} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

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